

Amrit Gopinath

Third-Year B.E. CSE Undergraduate • NLP, LLM Interpretability, and Multimodal Research

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EDUCATION

Sri Sivasubramaniya Nadar College of Engineering, Chennai

2024–Present

B.E. Computer Science and Engineering | CGPA: **9.383/10.0** | Department Rank: **1st**

RESEARCH FOCUS

Multilingual NLP and representation analysis; language-model interpretability and model introspection; mixture-of-experts routing; emerging work on vision–language models.

SELECTED RESEARCH

A Declarative–Procedural Perspective on Expert Routing in Bilingual MoE Language Models

2026

ARR Submission | [arXiv Preprint](#) | [First Author](#)

- Asked whether expert routing in bilingual English–German MoE language models primarily reflects language identity and lexical information or higher-level grammatical and syntactic structure.
- Designed controlled bilingual corpora, curriculum and no-curriculum training schedules, and held-out lexical, grammatical, and syntactic probes to separate these explanations.
- Analyzed routing with entropy, mutual and conditional mutual information, Jensen–Shannon divergence, frequency controls, and targeted routing ablations.
- Led the research question, experimental design, end-to-end analysis pipeline, and manuscript preparation.

Having a State Is Not Knowing It

2026

[Project](#) | [Report](#) | [Apart Research Sprint](#) | [Co-author](#)

- Investigated whether language models can report causally relevant internal decision states, separating possession of a state from verbal access to it through a falsifiable hierarchy of introspection capabilities.
- Replicated concept-injection experiments, then built a controlled Qwen benchmark using measured logit-margin shifts as ground truth for self-report, external decoding, and counterfactual intervention prediction.
- Native self-report reached **0.24 F1**, trainable self-report **0.52 F1**, and external probes decoded the state perfectly. Models predicted intervention direction at **0.70 F1** but not magnitude, separating qualitative state sensitivity from quantitative self-modeling.

RESEARCH EXPERIENCE

NLP Research Intern

2026

National Institute of Technology Tiruchirappalli | Submitted to ICON 2026 | [Paper](#) | [Code](#)

- Under faculty mentorship, investigated computational modeling of Classical Tamil *urai*, focusing on what linguistic structure and learned representations can recover in a low-resource setting.
- Developed mBART-style encoder-decoder and decoder-only generation/evaluation pipelines, alongside syntax-aware tags and explicit linguistic features as candidate inductive biases.
- Tested out-of-domain generalization on Thirukural couplets and analyzed clause structure, predicates, finite and non-finite constructions, and morphosyntactic signals for retrieval and representation learning.

SUPPORTING MULTIMODAL AND BENCHMARK WORK

Exploring Swin Transformers, Luminance Input, and Hybrid Architectures for Synthetic Image Detection

2026

MediaEval 2026 Working Notes | [Paper](#) | [Code](#) | [First Author](#)

- Compared Swin, luminance-only, CNN-sequence, and hybrid architectures to test whether luminance and localized patch evidence improve robustness across synthetic-image generators; the strongest Swin system achieved **0.6997 F1**.

FinMMEval 2026 Shared Task Submissions – TextSentinels

2026

CLEF 2026 Working Notes – Accepted | [Papers: T1, T2, T3](#) | [Co-author](#)

- Built multilingual retrieval, grounded financial QA, and decision-making systems spanning FAISS model routing, BM25 over SEC filings and tables, and sentiment-aware trading; ranked **6th/6th/7th** in English/Chinese/Arabic MCQA.
- Improved local financial-QA ROUGE-1 F1 from **0.04 to 0.28**; the forward-tested trading system reached **1.14 cumulative return** and **2.03 Sharpe ratio**.

TECHNICAL SKILLS

ML / NLP	PyTorch, Hugging Face Transformers, scikit-learn, FAISS, Stanza
Analysis Methods	Mutual and conditional mutual information, routing entropy, Jensen-Shannon divergence, permutation tests, CCA, concept injection, external probes
Programming / Tools	Python, C, Java, TypeScript, Git, Linux, LaTeX

ACADEMIC STANDING AND SERVICE

- **Merit Scholarship:** 3rd Position, First Year.
- **Reviewer / Sub-reviewer, CLEF 2026 Working Notes:** reviewed methodological clarity, experimental validity, presentation, and reproducibility.